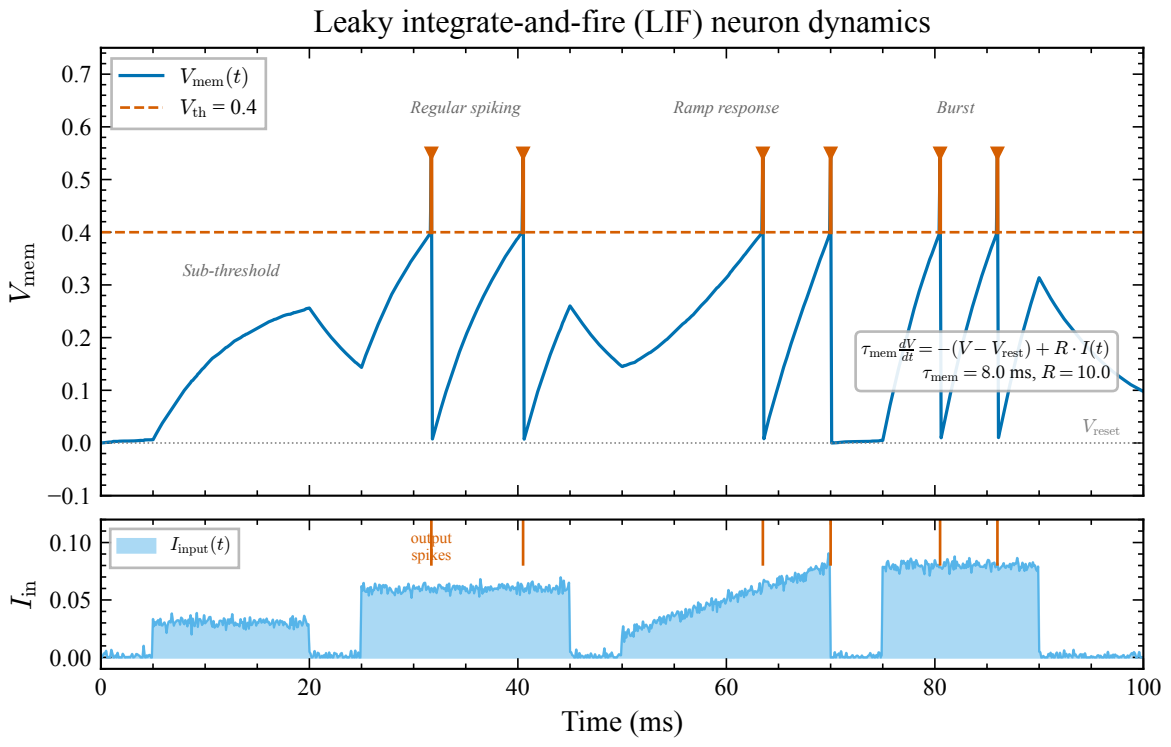

Leaky integrate-and-fire (LIF) neuron dynamics
$V_{\text{mem}}(t)$
$V_{\text{th}} = 0.4$
Regular spiking
Ramp response
Burst
Sub-threshold
$\tau_{\text{mem}} \frac{dV}{dt} = -(V - V_{\text{rest}}) + R \cdot I(t)$
$\tau_{\text{mem}} = 8.0$ ms, $R = 10.0$
V_{reset}
V_{mem}
$I_{\text{input}}(t)$
output spikes
I_{in}
Time (ms)